MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT ART, DESIGN & TECHNOLOGY UNIVERSITY
PUNE, INDIA
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Subject: IDS Roll No.: 43 Exp. No.: 1

Name of the Experiment:

Performed on:

Submitted on:

Marks	Teacher's Signature with date
10	

Q1 Define an algorithm list any 4 charatoris tics of algorithm

Soln A Step by Step produce or formula for solving a Problem is known as alogorithm

Characteristics are

1 Well defined input : It Should Procedure accept zero or more inputs.

2 Well defined output : It Should Produce at least one output

3 finiteness : The algorithm must terminate after a no. of steps.

4 Definiteness : each step must clear and unambiguous.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Q2] Define data structure and Abstract data type (ADT). give one exampl of each.

→ A data structure is a way of organizing and storing data so it can be used efficiently.

example :-
Array :- Stores element in contiguous memory loction.

Abstract Data type.
An ADT is logical Model that defines what operation can be Performed on data, without Specifying how they are implemented

Example :-
Stack ADT - operation : push, pop, peak.

Q3] Explain the Process of converting a Problem into Program. Illustrate the role of algorithms.

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Process.

1. Problem definition - Understan the Problem clearly
2. Analysis - Identify input, output and constraints.
3. Algorithm design - Create step-by-S[tep] Solution.
4. Flowchart / Pseudocode - represent logic visually.
5. Coding - Convert logic into Program.
6. Testing & debugging - Find and fix errors.
7. Documentation & Maintenance - Record and update.

Role of algorithm.

Algorithm act as a blueprint for the Program the ensure logical Correctness before Coding. They in writting efficient.

Q4) Explain linear and non-linear Structure data structures with example.

- linear Data Structures: Data elements are arranged sequentially.
Ex: Arrays, linked list, Stack, Queue.

- Non linear data Structures.
Data elements are not stored sequentially they form hierarchial or network relationships.
Ex = Tree, graph.

Q5) Differentiate between and algorithm, Analyse their roles in SDLC

Algorithm	Program
Step by-Step logic	actual Code written in a language
language independent	language dependent

Focus on Solution	Focus on implementation
Written before coding	Written during Coding

Roles in SDLC.

1. Algorithms used in design phase plan solution

2. Program used in implementation phase to execute.

3. Good algorithm lead to efficient Program

Q6] Relationship among data, data structure and algorithm.

Data, Raw facts or values.
Data structure organises data
Algorithm processes data

effect on efficiency
Speed
Memory use.

Examples : searching in an array (linear search)

Searching in a binary search Tree → faster.